Question 6: Short Answer — 4 points

A For the correct equation (state symbols not required): **Point 01**

$$Al(s) \rightarrow Al^{3+}(aq) + 3\,e^-$$

B For the correct balanced net ionic equation (state symbols not required): **Point 02**

$$2\,Al(s) + 3\,Zn^{2+}(aq) \rightarrow 2\,Al^{3+}(aq) + 3\,Zn(s)$$

C For the correct answer and a valid justification that correctly compares the masses of Al and Zn based on their molar masses and the stoichiometry of the balanced equation. **Point 03**

Examples of acceptable responses may include the following:

- Zn *experiences a greater change in mass. Assuming the entire* Al *anode reacts:*

  $$50.0\text{ g Al} \times \frac{1\text{ mol Al}}{26.98\text{ g Al}} \times \frac{3\text{ mol Zn}}{2\text{ mol Al}} \times \frac{65.38\text{ g Zn}}{1\text{ mol Zn}} = 182\text{ g Zn}$$

- Zn *experiences a greater change in mass.*

  $$1\text{ mol}_{rxn} \times \frac{3\text{ mol Zn}}{1\text{ mol}_{rxn}} \times \frac{65.38\text{ g Zn}}{1\text{ mol Zn}} = 196.1\text{ g Zn}$$

  $$1\text{ mol}_{rxn} \times \frac{2\text{ mol Al}}{1\text{ mol}_{rxn}} \times \frac{26.98\text{ g Al}}{1\text{ mol Al}} = 53.96\text{ g Al}$$

  Thus, for however many moles of reaction that proceed, the mass of Zn *produced will be greater than the mass of* Al *consumed.*

- Zn *experiences a greater change in mass. As the reaction proceeds, three moles of* Zn *are used for every two moles of* Al. *Thus, for every* 196 g *of* Zn *that are produced,* 54 g *of* Al *are consumed.*

D For the correct calculated value. **Point 04**

Examples of acceptable responses may include the following:

- $E^\circ_{cell} = 1.50\text{ V} + 0.76\text{ V} = 2.26\text{ V}$
- $Au^{3+}(aq) + 3\,e^- \rightarrow Au(s) \qquad +1.50\text{ V}$

  $Zn(s) \rightarrow Zn^{2+}(aq) + 2\,e^- \qquad +0.76\text{ V}$

  $E^\circ_{cell} = 2.26\text{ V}$

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